1. Computer Architectures

The Management of the Information

(Additional Resources)

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Two’s Complement

Two's Complement is a method used in computer science and electronics to represent **signed integers** in binary systems. It is particularly useful because it **simplifies arithmetic operations**, such as addition and subtraction, allowing both **positive** and **negative numbers** to be handled by the same logical circuits.

Basics Concepts:

- The most significant bit (MSB) is used as the sign bit:
  - 0: indicates a positive number.
  - 1: indicates a negative number.
- **Negative numbers** are represented by **calculating the two's complement** of their corresponding **positive value**. **Positive numbers** are represented by **calculating the two's complement** of their corresponding **negative value**.

Examples:

+3 in 4-bit binary is $\boxed{0}$ 011 ⟶ positive number

MSB

-3 in 4-bit binary is $\boxed{1}$ 101 ⟶ negative number

MSB

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How to Calculate Two's Complement 1/2

To find the two's complement of a binary number:

1. **Invert all the bits** of the binary number (one's complement).
2. **Add 1** to the result of the previous step.

Positive -> Negative

Let's assume we are using 4-bit numbers.

Representation of +3:

- The binary representation is: 0011 (4 bits).

Representation of -3:

- **Invert the bits** of +3 (0011): the result is 1100 (one's complement).
- **Add 1 (i.e., 0001)** to 1100.

```
   1100
 + 0001
 -------------
   1101 ⟶ -3 in 4-bit binary
```

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How to Calculate Two's Complement 2/2

To find the two's complement of a binary number:

1. **Invert all the bits** of the binary number (one's complement).
2. **Add 1** to the result of the previous step.

Negative -> Positive

Let's assume we are using 4-bit numbers.

Representation of -3:

- The binary representation is: 1101 (4 bits).

Representation of +3:

- **Invert the bits** of -3 (1101): the result is 0010 (one's complement).
- **Add 1 (i.e., 0001)** to 0010.

```
   0010
 + 0001
 -------------
   0011 ⟶ +3 in 4-bit binary
```

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Two’s Complement Properties

- **Single Zero Representation:** In two's complement, positive zero (`0000`) and negative zero (`0000`) share the same binary representation, avoiding ambiguity.

- **Ease of Addition and Subtraction:** To subtract a number, you simply add its two's complement.

  Examples:

  - “5 – 3” can be performed as “5 + (– 3)” in binary.
  - “-5 – 3” can be performed as “-5 + (-3)” or as “(5 + 3) applying the two's complement to the result” in binary.

  By converting subtraction into addition using two's complement, binary arithmetic becomes simpler and more efficient, particularly in digital systems like computers and calculators.

- **Range of Values:** For an N-bit system, two's complement allows representation of numbers from -2^{N-1} to $2^{N-1} - 1$. For example, with 4 bits, from -8 to +7.

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